Maintenance Advisory - 10 MVA, 33/11 kV Power Transformer

As of 2018-06-26 | To: Head of Maintenance | From: Asset Analytics

Health index	Band	Urgency
45.0 / 100	Poor	HIGH - within 2 weeks

Component scores: **thermal** 82.9, **loading** 85.0, **oil** 65.0, **dga** 58.0

Top risk

Thermal fault 300-700 C (ML: T2, Duval Triangle 1: T2)
Drivers: Buchholz Alarm (cap) = True (-22.0 pts); Moisture (ppm) = 34.6 (-4.6 pts); CH4 (ppm) = 446.4 (-4.4 pts)

Recommended action

Typical of bad joints, circulating currents in core/clamps or OLTC contacts: confirm with lab DGA, IR thermography of bushings/OLTC, winding resistance + core-insulation (megger) test at the next planned outage; cap loading at 0.8 pu until resolved. Buchholz gas alarm in last 30 days: sample Buchholz gas and analyse with oil DGA before any re-energisation after trip.

Cost-benefit (12-month expected cost)

Item	INR
Cost if failure occurs (repair + outage)	INR 3.48 crore
Probability of failure, 12 months (assumed for band)	12%
Expected loss - no action	INR 41.8 lakh
Cost of recommended action	INR 13.6 lakh
Expected loss - after action	INR 12.5 lakh
Net benefit of acting now	**INR 15.7 lakh (B/C 2.2)**

Insulation ageing (IEC 60076-7): 1.79 % of normal life used in the monitored period; max hot-spot 137 C.
Events last 30 days: BUCHHOLZ_ALARM, WTI_ALARM.

Prototype built on public (ETT, IEC TC 10) and synthetic data. Costs and failure probabilities are stated assumptions (config.yaml). Confirm with laboratory DGA and site inspection before acting.